

Efficacy and safety of neoadjuvant stereotactic body radiation therapy plus dalpiciclib and exemestane for hormone receptor-positive, HER2-negative breast cancer: A prospective pilot study

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Yu Zhang, Shuo Cao, Nan Niu, Huilian Shan, Jinqi Xue, Guanglei Chen, Yongqing Xu, Jianqiao Yin, Chao Liu, Lisha Sun, Xiaofan Jiang, Meiyue Tang, Qianshi Xu, Mingxuan Jia, Xu Zhang, Zhenyong Zhang, Qingfu Zhang, Jianfei Wang, Ailin Li, Yongliang Yang, Caigang Liu ✉

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China • Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical University, Shenyang, China • Innovative Cancer Drug Research and Development Engineering Center of Liaoning Province, Shenyang, China • Department of Gastrointestinal Surgery, Yantai Affiliated Hospital of Binzhou Medical University, Yantai, China • Department of Pathology, the First Affiliated Hospital and College of Basic Medical Sciences of China Medical University, Shenyang, China • Jiangsu Hengrui Pharmaceuticals, Shanghai, China • School of Bioengineering, Dalian University of Technology, Dalian, China

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eLife Assessment

The study evaluates the feasibility, safety, and tolerability of neoadjuvant radiotherapy followed by a CDK4/6 inhibitor (dalpiciclib) and hormonal therapy in treatment-naïve patients with unilateral early-stage HR+/HER2- breast cancer. The findings are **convincing**, with a strong scientific rationale supported by integrated correlative studies. The trial is considered to be **important** as the outcomes could inform the design of larger, future studies. The limitations of the study have been acknowledged and outlined in this manuscript, which include only a small cohort of patients (n=12), which was not adequately powered to definitively assess the efficacy or safety of this combinatorial treatment approach.

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Abstract

Background

Both neoadjuvant chemotherapy and endocrine therapy only result in trivial pathological complete response rates and moderate objective response rates (ORR) in hormone receptor (HR)-positive, human epidermal growth factor receptor-2 (HER2)-negative breast cancer,

more promising alternatives are urgently needed. With proven synergistic effect of cyclin dependent kinase 4/6 (CDK4/6) inhibitor and radiotherapy in preclinical studies, this pilot study aimed to explore the efficacy and safety of neoadjuvant stereotactic body radiation therapy (SBRT) followed by dalpiciclib and exemestane in HR-positive, HER2-negative breast cancer.

Methods

This was a single-arm, non-controlled prospective pilot study. Treatment-naïve patients with unilateral HR-positive, HER2-negative breast cancer received neoadjuvant radiotherapy (24Gy/3F) followed by dalpiciclib and exemestane for 6 cycles. The primary endpoint was the proportion of patients with residual cancer burden (RCB) score of 0-I. Key secondary endpoints included ORR, breast-conservation rate, biomarker analysis, and safety.

Results

All 12 enrolled patients completed the study treatment and surgery. Two (16.7%) of them achieved the RCB 0-I with the ORR of 91.7% (11/12). Analyses of tumor specimens showed significant increase of infiltrating T cells rather than alteration of PD-L1 positive immune cells. The most common grade 3 adverse events (AEs) were neutropenia (66.7%) and leukopenia (25.0%), but no grade 4-5 AE or death occurred.

Conclusions

Our results suggested neoadjuvant SBRT followed by dalpiciclib and exemestane are effective and tolerable, and provides novel insights for the neoadjuvant treatment of HR+/HER2-breast cancer, which may be considered as a feasible option for patients with HR-positive, HER2-negative breast cancer.

Impact statement

Neoadjuvant SBRT with dalpiciclib and exemestane resulted in a 16.7% RCB 0-I rate and a 91.7% ORR, with acceptable toxicities in HR-positive, HER2-negative breast cancer patients.

Introduction

Hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative breast cancer is the most prevalent subtype of breast cancer, which accounts for approximately 70% of all breast cancer (Parker et al., 2009 [\[1\]](#)). Neoadjuvant chemotherapy (NCT) is recommended as the first-choice for patients with HR-positive, HER2-negative breast cancer, who are candidates for preoperative therapy, according to National Comprehensive Cancer Network (NCCN) guidelines(Gradishar et al., 2022 [\[2\]](#)). Neoadjuvant endocrine therapy (NET) is an acceptable, less toxic alternative of NCT. However, less than 10% of patients with HR-positive, HER2-negative breast cancer can achieve pathological complete response (pCR), and only 60% of patients respond to NCT or NET (Sella et al., 2021 [\[3\]](#)). Hence, more promising alternatives are urgently needed.

Cyclin dependent kinase 4/6 (CDK4/6) inhibitors are breakthroughs in the treatment of HR-positive, HER2-negative breast cancer, and have a remarkable efficacy in the advanced setting when combined with endocrine therapy (ET) (Finn et al., 2015 [↗](#); Goetz et al., 2017 [↗](#); Hortobagyi et al., 2016 [↗](#)). Presently, abemaciclib has been approved for high-risk, early HR-positive breast cancer plus ET as adjuvant treatment and its application is being explored in neoadjuvant setting (neoMONARCH and NCT04293393). However, therapies with combination of CDK4/6 inhibitors and NET only achieve a pCR rate of less than 5% (Hurvitz et al., 2020 [↗](#); Johnston et al., 2019 [↗](#)). In the PALLET study, objective response rate (ORR) was not statistically significantly different between the combinational therapy and monotherapy (54.3% vs 49.5%, $P = 0.20$) (Johnston et al., 2019 [↗](#)). Likewise, in the FELINE study, there was no statistical difference in clinical, mammographic, ultrasound or magnetic resonance imaging (MRI) response between letrozole alone and combination of letrozole with a CDK4/6 inhibitor (Khan et al., 2020 [↗](#)). Thus, it is reasonable to explore a novel neoadjuvant therapy for HR-positive, HER2-negative breast cancer.

Radiotherapy is one of the most important local methods to control malignancy by breaking DNA in tumor cells (Cox & Swanson, 2013 [↗](#)). Stereotactic body radiation therapy (SBRT) is a novel and relatively safe radiotherapy technique and delivers high doses of radiation in a small number of fractions to the tumor lesion while minimizing radiation exposure to the surrounding tissues (Chmura et al., 2021 [↗](#)). Currently, neoadjuvant radiotherapy can be applicable for inoperable breast cancer to allow an effective and sometimes more conservative surgery (Cardoso et al., 2018 [↗](#)). Preclinical studies have indicated that CDK4/6 inhibitors can sensitize breast cancer cells to radiotherapy, possibly by inhibition of DNA damage repair response, enhancement of apoptosis and blockage of cell cycle progression, and inhibition of intra-tumor cellular metabolism and angiogenesis (Han et al., 2013 [↗](#); A. M. Pesch et al., 2020 [↗](#); G. Petroni et al., 2021 [↗](#)). A preclinical study has explored different schedules of combination of radiotherapy and CDK4/6 inhibitors in human and mouse model and demonstrated that radiotherapy followed by CDK4/6 inhibitors can enhance antineoplastic effects, compared with their monotherapy alone or CDK4/6 inhibitors concurrent with or followed by radiotherapy (G. Petroni et al., 2021 [↗](#)). Moreover, the combination of CDK4/6 inhibitors and radiotherapy can lead to a satisfying overall disease control and safety based on several retrospective studies in metastatic breast cancer (Bosacki et al., 2021 [↗](#)). Nevertheless, the efficacy and safety of this regimen in neoadjuvant treatment of early or locally advanced HR-positive, HER2-negative breast cancer are unknown. We have conducted a prospective pilot study to determine the efficacy and safety of neoadjuvant radiotherapy followed by dalpiciclib and exemestane in patients with newly-diagnosed early or locally advanced HR-positive, HER2-negative breast cancer. Dalpiciclib (SHR6390) is a potent selective inhibitor of CDK4/6 and exerts synergistic anti-tumor effect when combined with endocrine therapy for advanced luminal breast cancer in the DAWNA-1 trials (Xu et al., 2021 [↗](#)).

Methods

Study design

This single-arm, non-controlled prospective pilot study in women with HR-positive, HER2-negative early or locally advanced breast cancer was registered with *ClinicalTrials.gov* [↗](#) (NCT05132790).

The study was approved by the Institutional Review Board and Ethics Committee of Shengjing Hospital of China Medical University, according to the Declaration of Helsinki and Good Clinical Practice guidelines. Written informed consent was obtained from each patient.

Participants

Eligible patients included pre- and post-menopausal women aged between 18 and 75 years, with histologically confirmed HR-positive, HER2-negative (0 or 1+ by immunohistochemistry [IHC], or 2+ by IHC with negative results of fluorescence in situ hybridization), early or locally advanced

invasive breast cancer with the lesion of ≥ 2 cm by MRI. Other main inclusion criteria were Eastern Cooperative Oncology Group performance status 0-1, adequate marrow, hepatic and renal function. The key exclusion criteria included stage IV tumors, inflammatory breast cancer, prior systemic treatment of the current cancer, other malignancies, cardiac disease or history of cardiac dysfunction, pregnancy, lactation, and refusal to use contraception.

Procedures

SBRT was performed every other day in breast target lesion with a total radiation dose of 24 Gy separated by 3 fractions. Within 2-4 weeks after the completion of radiotherapy, individual patients received 6 cycles of combination of oral 150 mg Dapiciclib daily on days 1-21 of each 4-week cycle and 25 mg exemestane with a total duration of 24 weeks. Patients without menopause were also injected subcutaneously with 3.6 mg goserelin every 4 weeks. All patients received study treatment until completion of all prescribed protocol therapy, disease progression, unacceptable toxicity, or withdrawal of consent. If disease progression, the patient was either proceed to surgery or received alternative neoadjuvant therapy. Surgery was performed 3-5 weeks after the last dose of dapiciclib to allow adverse event (AE) recovery. Recommended surgery and adjuvant therapy followed per local guidelines or institutional standards. The necessity of additional chemotherapy for patients with pCR was judged by investigators. A 5-year follow-up is conducted every 3 months during the first 2 years, and every 6 months for the subsequent 3 years. Additionally, safety data are collected within 90 days after surgery for subjects who discontinue study treatment.

Outcomes

The primary endpoint was the rate of residual cancer burden (RCB) 0-I. Secondary endpoints included ORR (defined as the proportion of patients with complete or partial response according to the Response Evaluation Criteria In Solid Tumors [RECIST] version 1.1 by MRI), pCR rate in the breast and axillary lymph nodes (tpCR; ypT0/is ypN0) and in the breast (bpCR; ypT0/is ypNx), breast-conservation rate, preoperative endocrine prognostic index (PEPI) score for breast cancer-specific survival (BCSS) (PEPI 0), and safety, according to the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

Patient-Reported Outcome (PRO) assessments

In the study, the patient-reported outcome (PRO) assessments was measured using the European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire Core-30 (QLQ-C30; version 3) (Cull, 1998 [\[4\]](#)) at baseline, at the end of cycles 1 to 6, and before surgery. The PRO data were scored, according to the EORTC scoring manual. A raw score was calculated as the average of items contributing to a scale and standardized to a range of 0 to 100 points. On global health status and functioning scales, a higher score indicated better status. Higher scores on symptom scales indicated worse symptoms.

Tumor-infiltrating lymphocytes (TILs)

Tumor samples were collected when performing needle biopsy and surgery. The specimens were fixed in 10% formalin and paraffin-embedded. The tissue sections (3 μ m) were stained with hematoxylin and eosin (H&E) and the frequency of TILs were quantified manually by a pathologist in a blinded manner. The TILs scores were determined as the percentage of TILs in tumor area (tumor cells and stroma). The percentage of stromal TILs (sTILs) was calculated in accordance with the recommendations by an International TILs Working Group 2014 (Salgado et al., 2015 [\[5\]](#)).

Programmed cell death-ligand 1 (PD-L1) expression

The expression of programmed cell death-ligand 1 (PD-L1) was characterized by immunohistochemistry (IHC) using anti-PD-L1 (clone SP142, Spring Bioscience, USA) and the percentages of PD-L1 positive tumor cells (TC) or infiltrating inflammatory cells (IC) were

quantified.

Statistical analysis

This exploratory study involves 12 patients, with the sample size determined based on clinical considerations, not statistical factors.

All statistical analyses were conducted using SAS 9.4 (North Carolina, USA). Continuous data are presented as mean and standard deviation (SD) or mean and 95% confidence interval (CI). Categorical data are expressed as frequency and percentage. The 95% CIs of pathological complete response rate, proportion of patients with RCB-0 or RCB-I, and ORR were estimated using the Clopper-Pearson method.

Results

Patient characteristics

Between November 2021 and January 2022, 12 HR-positive, HER2-negative patients were screened and recruited in the trial. Their baseline demographic and clinical characteristics are shown in **Table 1** [↗](#). They had a median age of 48.5 years (range, 36-56) and the median tumor size of 33.5mm (range, 23-90) assessed by breast MRI.

Patient outcomes

After the completion of study treatment, 2 (16.7%, 95% CI 3.5%-46.0%) out of 12 patients achieved RCB 0-I tumor response, tpCR and PEPI 0 (**Table 2** [↗](#)). There were 3 (25.0%) patients reached bpCR. Overall, this study led to an ORR (as assessed by breast MRI) of 91.7% (95% CI 62.5%-100%, **Fig. 1** [↗](#)). Rate of conversion from mastectomy at baseline to breast conservation at surgery was 25.0% (95% CI 8.3%-53.9%). The median Ki67 of surgical specimens for patients without pCR was 7% (range, 5-70).

Toxicity

All patients completed the study treatment and AEs are summarized in **Table 3** [↗](#). The most frequent grade 1-2 AEs were leukopenia (58.3%), hot flushes (50.0%) and lymphopenia (50.0%). There were some patients with Grade 3 AEs, including neutropenia (66.7%) and leukopenia (25.0%), but not grade 4 AEs. There was no serious AEs, no therapy-related death and no patients with dose interruption and reduction or treatment discontinuation in this population. Furthermore, there was no patient with radiation-related dermatitis and skin hyperpigmentation in this population.

Patient-reported outcome

All patients completed the EORTC QLQ-C30 and had available results of PRO assessment (**Fig. 2** [↗](#)). There was no clinically significant deterioration in global health status, physical, role, emotional, cognitive and social functioning in this population.

Alteration in tumor immune microenvironment

To examine the impact of this new neoadjuvant therapy on tumor environment, we performed H&E staining of tissue sections in 10 evaluable patients (**Fig. 3** [↗](#)). Compared with those in pre-treatment biopsied tissues, treatment significantly increased the percentages of tumor-infiltrating

	Patients (n=12)
Median age (range), years	48.5 (36-56)
≤ 50	6 (50.0%)
> 50	6 (50.0%)
Menopausal status	
Premenopausal	6 (50.0%)
Postmenopausal	6 (50.0%)
Tumor size	
T2	11 (91.7%)
T3	1 (8.3%)
Lymph node status	
N0	5 (41.7%)
N1	2 (16.7%)
N2	5 (41.7%)
Clinical stage	
IIA	4 (33.3%)
IIB	3 (25.0%)
IIIA	5 (41.7%)
Tumor grade	
II	11 (91.7%)
III	1 (8.3%)
ER expression	
> 10%	12 (100.0%)
PgR expression	
< 20%	3 (25.0%)
≥ 20%	9 (75.0%)
HER2 expression	
0	7 (58.3%)
1+	1 (8.3%)
2+, FISH	4 (33.3%)
Median baseline Ki67 (range), %	35 (15-70)
≤ 14%	3 (25.0%)
> 14%	9 (75.0%)

Data are n (%) or median (range).

ER: estrogen receptor; PgR: progesterone receptor; HER2: human epidermal growth factor receptor 2; FISH: fluorescence in situ hybridization.

Table 1

Baseline characteristics of the patients.

Figure 1

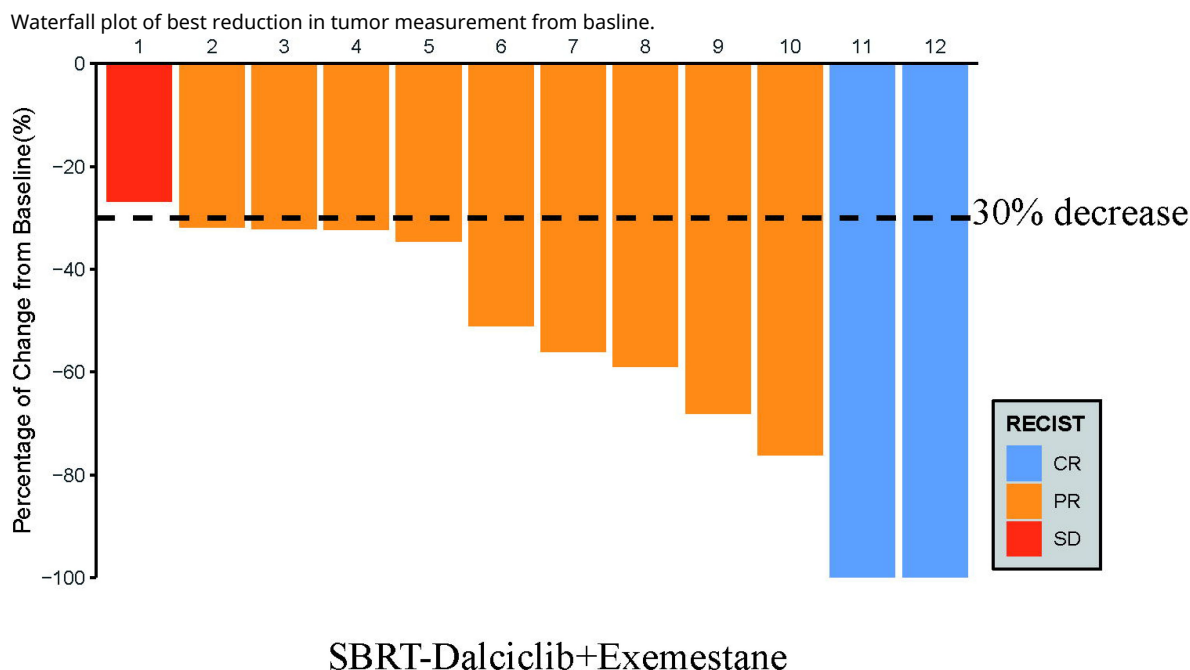


Table 2

Pathological and clinical response (n=12).

	n (%)
Total pathological complete response	2 (16.7%)
Breast pathological complete response	3 (25.0%)
Residual cancer burden score	
0	2 (16.7%)
I	0
II	6 (50.0%)
III	4 (33.3%)
Radiological response	
Complete response	2 (16.7%)
Partial response	9 (75.0%)
Stable disease	1 (8.3%)
Objective response rate	11 (91.7%)
Preoperative endocrine prognostic index	
0	2 (16.7%)
1-3	0
≥4	10 (83.3%)

Data are presented as n (%)

	n (%)		
	Grade 1 or 2	Grade 3	Grade 4
Neutropenia	3 (25.0%)	8 (66.7%)	0
Leukopenia	7 (58.3%)	3 (25.0%)	0
Hot flushes	6 (50.0%)	0	0
Lymphopenia	6 (50.0%)	0	0
Insomnia	4 (33.3%)	0	0
Alopecia	4 (33.3%)	0	0
Anemia	4 (33.3%)	0	0
Hyperuricemia	4 (33.3%)	0	0
Blood urea increased	4 (33.3%)	0	0
Rash	3 (25.0%)	0	0
Thrombocytopenia	3 (25.0%)	0	0
Hyperglycemia	3 (25.0%)	0	0
Creatinine increased	3 (25.0%)	0	0
Fatigue	3 (25.0%)	0	0
Arthralgia	2 (16.7%)	0	0
γ-glutamyl transferase increased	2 (16.7%)	0	0
Hypocalcemia	2 (16.7%)	0	0
Abnormal T wave of electrocardiogram	2 (16.7%)	0	0
Hyponatremia	2 (16.7%)	0	0
Lactate dehydrogenase increased	2 (16.7%)	0	0
Diarrhea	1 (8.3%)	0	0
Headache	1 (8.3%)	0	0
Mucosal inflammation Constipation	1 (8.3%)	0	0
Proteinuria	1 (8.3%)	0	0
Hypertriglyceridemia	1 (8.3%)	0	0
Alanine aminotransferase increased	1 (8.3%)	0	0

Data are presented as n (%).

Table 3

Treatment-emergent adverse events (n=12).

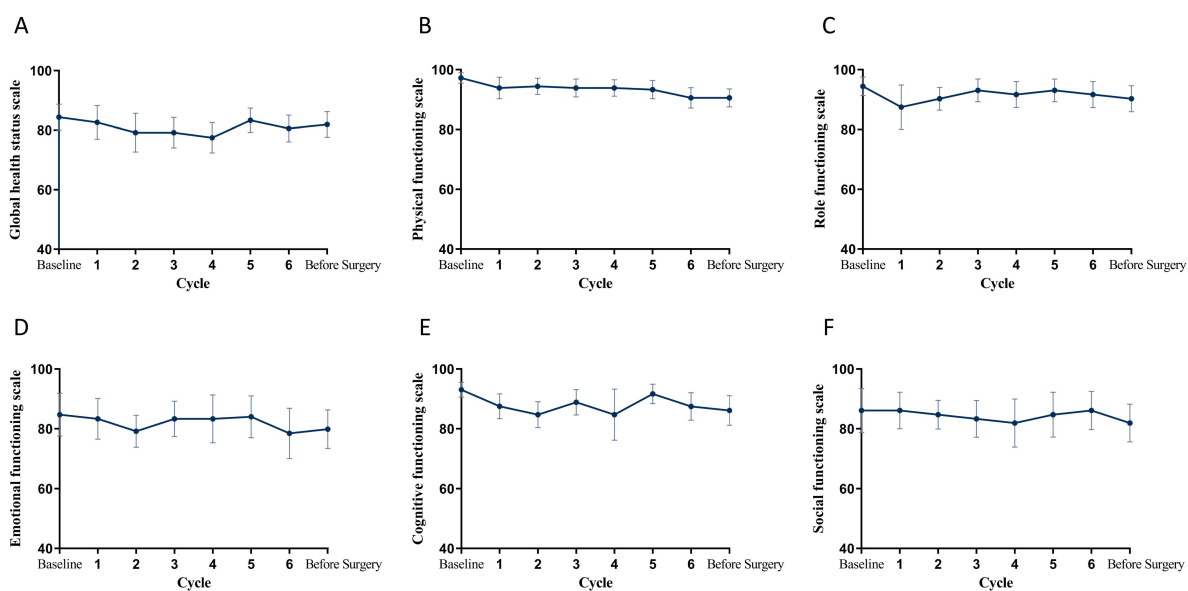


Figure 2

Mean change from baseline in QLQ-C30 over time.

(A) Global health status. (B-F) Physical, role, emotional, cognitive and social functioning.

lymphocytes (TILs) in the tumor environment (22.9% vs. 10.7%, $P=0.043$). IHC staining of PD-L1 revealed that there appeared no significant difference in the percent of PD-L1 positive immune cells and tumor cells between the pre-treatment and post-treatment specimens.

Discussion

This pioneering study reported the efficacy and safety of sequential neoadjuvant radiotherapy and combination of dalpiciclib and exemestane in patients with HR-positive, HER2-negative breast cancer. This new therapeutic regimen achieved considerable anti-tumor activity with the RCB 0-I rate of 16.7% and ORR rate of 91.7%, respectively. More importantly, this combinatory therapeutic strategy was well tolerated in this population.

CDK4/6 inhibitors have recently been preclinically identified as radiosensitizers, acting mainly through hindering DNA damage repair response after radiation exposure, and enhancing cell apoptosis and blocking cell cycle, which provide a conceptual basis of combining radiotherapy with a CDK4/6 inhibitor (Khasraw et al., 2017; Xie et al., 2019). Moreover, both radiotherapy and CDK4/6 inhibitors exert immunomodulatory effects and may synergize anticancer immune responses (Andrea M Pesch et al., 2020). SBRT with hypofractionation and higher radiation doses is more effective than conventional fractionation in triggering immune responses (Gandhi et al., 2015). Consistent with the basic research (Giulia Petroni et al., 2021), we used the sequential treatment with SBRT and combination of dalpiciclib and exemestane.

Unlike HER2-positive and triple-negative breast cancer, NCT or NET for HR-positive, HER2-negative breast cancer only achieves a pCR rate of $\leq 10\%$ (Cortazar et al., 2014). Our study indicated that this novel therapeutic regimen achieved a pCR and RCB 0-I rate of 16.7%, while neoadjuvant therapies with combination of a CDK4/6 inhibitor and letrozole led to a pCR rate of 0-3.8% and RCB 0-I rate of 6.1-7.7% and treatment with NCT resulted in a pCR rate of 5.8-5.9% and RCB 0-I rate of 11.8-15.7% in HR-positive, HER2-negative breast cancer patients (C. X. Ma et al., 2017; Prat et al., 2020). It is notable that pCR rate (commonly defined as ypT0/isN0) is not significantly associated with the prognosis of HR-positive, HER2-negative breast cancer (Cynthia X Ma et al., 2017). In addition, clinical trials to test different therapies for patients with HR-positive, HER2-negative breast cancer have focused on clinical response rates as important clinical endpoints (Hurvitz et al., 2020; Johnston et al., 2019; Khan et al., 2020; Khasraw et al., 2017; C. X. Ma et al., 2017). In our study, treatment with the new therapeutic regimen led to eleven (91.7%) patients achieving objective responses by MRI, while previous studies have revealed that NET achieves an objective response rate of approximately 60% in HR-positive, HER2-negative breast cancer patients (Sella et al., 2021). One of the principal objectives of neoadjuvant therapy is to facilitate downstaging and enhance the rate of breast conservation. With the addition of SBRT, a slightly higher rate of patients changed from mastectomy to breast conservation compared with PALLET study (25.0% vs 14.1%). (Johnston et al., 2019). Consequently, our new neoadjuvant therapeutic regimen appeared to improve the likelihood of pathological downstaging and achieve a margin-free resection, particularly for those with locally advanced and high-risk breast cancer.

Considering that treatment with radiotherapy or a CDK4/6 inhibitor can enhance anti-tumor immunity in vitro, We analyzed the impact of new neoadjuvant therapeutic regimen on alternation in biomarker expression in the numbers of TILs before and after the treatment. Our study showed a modest but statistically significant increase in the tumor tissues with the addition of SBRT. Apparently, sequential neoadjuvant therapies with radiotherapy, a CDK 4/6 inhibitor and exemestane turned non-responsive 'cold' tumors into responsive 'hot' ones. Therefore, our findings may provide significant insights into designing effective therapeutic strategies against HR-positive, HER2-negative breast cancer.

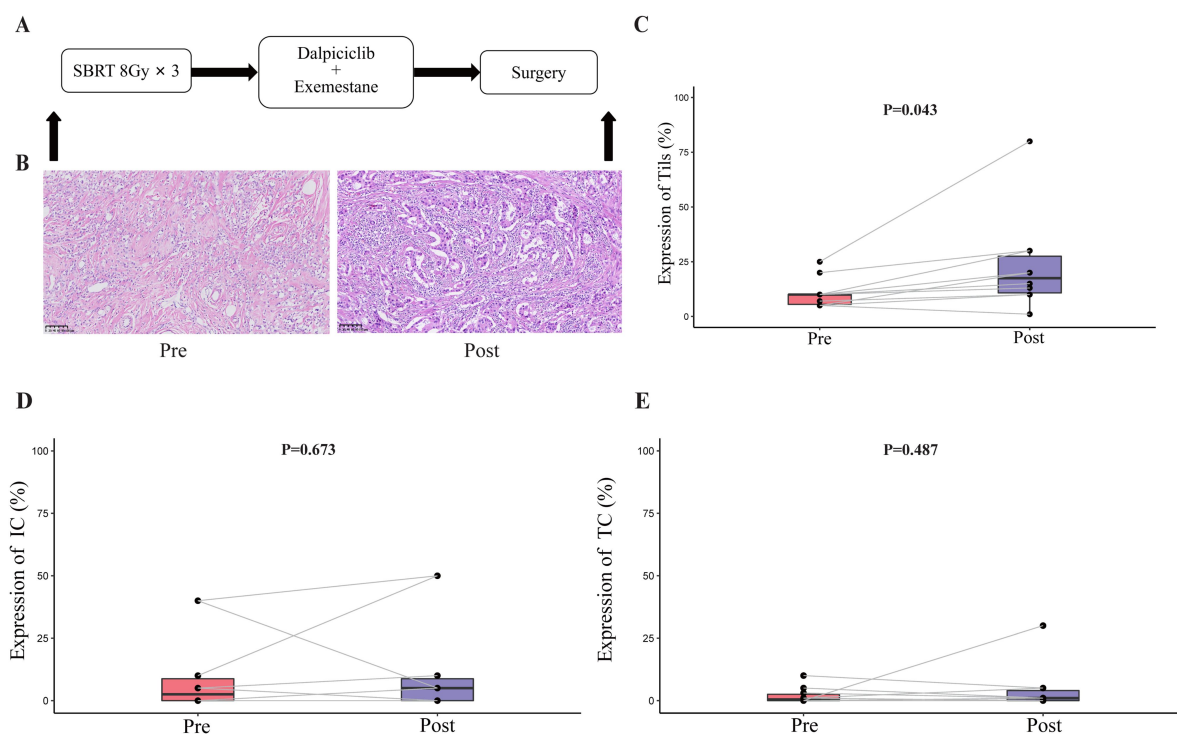


Figure 3

TILs and PD-L1 changes before and after treatment.

(A) Study schema. (B) Representative TILs. Scale bar, 100 μ M. (C-E) Percentage of TILs, PD-L1 expression in immune cell and tumor cell in paired specimens (n=10). Bars, boxes, and whiskers represent median, interquartile range, and range, respectively.

Patients with HR-positive, HER2-negative breast cancer were well-tolerated to the combinational regimen. During the observation period, those patients only developed grade 1-3 of AEs, including grade 3 neutropenia (66.7%), similar to that of combination of dalpiciclib and fulvestrant in the DAWNA-1 trial (84.2%) (Xu et al., 2021 [↗](#)). There was no new AEs following therapies with radiotherapy and dalpiciclib in this population.

We recognized that our study had limitations, including a one-arm preliminary exploratory trial with small sample size. The small sample size limited the comparison of our data with historical data in the literature due to the potential bias. Consequently, cross-trial comparisons like this hold limited significance. Thus, further prospective randomized clinical trials with a larger population for longer treatment duration are warranted to validate the findings.

In conclusion, our study provides novel insights into the neoadjuvant treatment of HR+/HER2-negative breast cancer. The sequential radiotherapy and a CDK4/6 inhibitor plus ET were effective and well-tolerated, which could serve as a feasible option for neoadjuvant therapy for HR-positive, HER2-negative breast cancers. Further validation of these findings is warranted in a large-scale study.

Data availability statement

The raw clinical and imaging data are protected due to patient privacy laws. The datasets generated and/or analyzed during the current study are available from the corresponding author Caigang Liu on request for 10 years; de-identified clinical data and experimental data are available on request sharing, which will need the approval of the Institutional Ethical Committees. De-identified data will then be transferred to the inquiring investigator over secure file transfer.

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Additional information

Clinical trial number: *NCT05132790* [↗](#).

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Author information

Yu Zhang

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China, Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering Center of Liaoning Province, Shenyang, China, Department of Gastrointestinal Surgery, Yantai Affiliated Hospital of Binzhou Medical University, Yantai, China

Shuo Cao

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Nan Niu

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Huilian Shan

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Jinqi Xue

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Guanglei Chen

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Yongqing Xu

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Jianqiao Yin

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Chao Liu

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Lisha Sun

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Xiaofan Jiang

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China

Meiyue Tang

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Qianshi Xu

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China

Mingxuan Jia

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Xu Zhang

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Zhenyong Zhang

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Qingfu Zhang

Department of Pathology, the First Affiliated Hospital and College of Basic Medical Sciences of
China Medical University, Shenyang, China

Jianfei Wang

Jiangsu Hengrui Pharmaceuticals, Shanghai, China

Ailin Li

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China

Yongliang Yang

School of Bioengineering, Dalian University of Technology, Dalian, China
ORCID iD: [0000-0003-0449-0599](https://orcid.org/0000-0003-0449-0599)

Caigang Liu

Department of Oncology, Shengjing Hospital of China Medical University, Shenyang, China,
Cancer Stem Cell and Translational Medicine Laboratory, Shengjing Hospital of China Medical
University, Shenyang, China, Innovative Cancer Drug Research and Development Engineering
Center of Liaoning Province, Shenyang, China
ORCID iD: [0000-0003-2083-235X](https://orcid.org/0000-0003-2083-235X)

For correspondence: liucg@sj-hospital.org

Editors

Reviewing Editor

Samir Zaidi

Yale School of Medicine, New Haven, United States of America

Senior Editor

Diane Harper

University of Michigan-Ann Arbor, Ann Arbor, United States of America

Reviewer #1 (Public review):

Summary:

This manuscript details the results of a small pilot study of neoadjuvant radiotherapy followed by combination treatment with hormone therapy and dalpiciclib for early stage HR+/HER2-negative breast cancer.

Strengths:

The strengths of the manuscript include the scientific rationale behind the approach, and the inclusion of some simple translational studies.

Weaknesses:

The main weakness of the manuscript is that a study this small is not powered to fully characterize efficacy or safety of a treatment approach, and can, at best, demonstrate feasibility. These data need validation in a larger cohort before they can have any implications for clinical practice, and the treatment approach outlined should not yet be considered a true alternative to standard evidence-based approaches.

I would urge the readers exercise caution when comparing results of this 12-patient pilot study to historical studies, many of which were much larger, and had different treatment protocols and baseline patient characteristics. Cross-trial comparisons like this are prone to mislead, even when comparing well powered studies. With such a small sample size, the risk of statistical error is very high, and comparisons like this have little meaning.

<https://doi.org/10.7554/eLife.101583.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1(Public review):

Summary:

This manuscript details the results of a small pilot study of neoadjuvant radiotherapy followed by combination treatment with hormone therapy and dalpiciclib for early-stage HR+/HER2-negative breast cancer.

Strengths:

The strengths of the manuscript include the scientific rationale behind the approach and the inclusion of some simple translational studies.

Weaknesses:

The main weakness of the manuscript is that overly strong conclusions are made by the authors based on a very small study of twelve patients. A study this small is not powered to fully characterize the efficacy or safety of a treatment approach, and can, at best, demonstrate feasibility. These data need validation in a larger cohort before they can have any implications for clinical practice, and the treatment approach outlined should not yet be considered a true alternative to standard evidence-based approaches.

I would urge the authors and readers to exercise caution when comparing results of this 12-patient pilot study to historical studies, many of which were much larger, and had different treatment protocols and baseline patient characteristics. Cross-trial comparisons like this are prone to mislead, even when comparing well powered studies.

With such a small sample size, the risk of statistical error is very high, and comparisons like this have little meaning.

We greatly appreciate your evaluation of our study and fully agree with the limitations you have pointed out. We have clearly stated the limitations of the small sample size and emphasized the need for a larger population to validate our preliminary findings in the discussion section (Lines 311-316).

We acknowledge that this small sample size is not powered to characterize this regimen as a promising alternative regimen in the treatment of patients with HR-positive, HER2-negative breast cancer. Therefore, we have revised the description of this regimen to serve as a feasible option for neoadjuvant therapy in HR-positive, HER2-negative breast cancers both in the discussion (Lines 317-320) and the abstract (Lines 71-72).

We agree with you that cross-trial comparisons should be approached with caution due to differences in study designs and patient populations. In our discussion section, we acknowledge that small sample size limited the comparison of our data with historical data in the literature due to the potential bias (Lines 312-313). We clearly state that such comparisons hold limited significance (Lines 313-314) and suggest a larger population to validate our preliminary findings.

• *Why was dalpiciclib chosen, as opposed to another CDK4/6 inhibitor?*

Thank you for your comments. The rationale for selecting dalpiciclib over other CDK4/6 inhibitors in our study is primarily based on the following considerations:

- (1) Clinical Efficacy: In several clinical trials, including DAWNA-1 and DAWNA-2, the combination of dalpiciclib with endocrine therapies such as fulvestrant, letrozole, or anastrozole has been shown to significantly extend the progression-free survival (PFS) in patients with hormone receptor-positive, HER2-negative advanced breast cancer [1-2].
- (2) Tolerability and Management of Adverse Reactions: The primary adverse reactions associated with dalpiciclib are neutropenia, leukopenia, and anemia. Despite these potential side effects, the majority of patients are able to tolerate them, and with proper monitoring and management, these reactions can be effectively mitigated [1-2].
- (3) Comparable pharmacodynamic with other CDK4/6 inhibitors: The combination of CDK4/6 inhibitors, including palbociclib, ribociclib, and abemaciclib, with aromatase inhibitors has demonstrated an enhanced ability to suppress tumor proliferation and increase the rate of clinical response in neoadjuvant therapy for HR-positive, HER2-negative breast cancer [3-5]. Furthermore, preclinical studies have shown that dalpiciclib has comparable in vivo and in vitro pharmacodynamic activity to palbociclib, suggesting its potential effectiveness in similar treatment regimens [6].
- (4) Accessibility and Regulatory Approval: Dalpiciclib has gained marketing approval in China on December 31, 2021, which facilitates the accessibility of this medication, making it a more convenient option when considering treatment plans.

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• *The eligibility criteria are not consistent throughout the manuscript, sometimes saying early breast cancer, other times saying stage II/III by MRI criteria.*

Thank you for pointing out the inconsistencies in the description of the eligibility criteria in our manuscript. We deeply apologize for any confusion caused by these inconsistencies. We have revised the term from “early-stage HR-positive, HER2-negative breast cancer” to “early or locally advanced HR-positive, HER2-negative breast cancer” (Lines 128 and 150). The term “early or locally advanced” encompasses two different stages of breast cancer, whereas “Stage II/III by MRI criteria” refers to specific stages within the TNM staging system.

• *The authors should emphasize the 25% rate of conversion from mastectomy to breast conservation and also report the type and nature of axillary lymph node surgery performed. As the authors note in the discussion section, rates of pathologic complete response/RCB scores are less prognostic for hormone-receptor-positive breast cancer than other subtypes, so one of the main rationales for neoadjuvant medical therapy is for surgical downstaging. This is a clinically relevant outcome.*

We appreciate your constructive comments. Based on your suggestions, we have made the following revisions and additions to the article.

The breast conservation rate serves as a secondary endpoint in our study (Line 62 and 179). We have highlighted the significant 25% conversion rate from mastectomy to breast conservation in both the results (Lines 229-230) and discussion sections (Lines 290-292).

In our study, all patients underwent lymph node surgery, including sentinel lymph node biopsy or axillary lymph node dissection. Among them, 58.3% of patients (7/12) underwent sentinel lymph node biopsies.

We agree with your point that the prognostic value of pathologic complete response/RCB score is lower for hormone receptor-positive breast cancer compared to other subtypes, we have revised the discussion section to clarify that one of the principal objectives for neoadjuvant therapy in this patient population is to facilitate downstaging and enhance the rate of breast conservation (Lines 289-290). And also emphasized that this neoadjuvant therapeutic regiment appeared to improve the likelihood of pathological downstaging and achieve a margin-free resection, particularly for those with locally advanced and high-risk breast cancer (Lines 293-295).

Reviewer #2 (Public review):

Firstly, as this is a single-arm preliminary study, we are curious about the order of radiotherapy and the endocrine therapy. Besides, considering the radiotherapy, we also concern about the recovery of the wound after the surgery and whether related data were collected.

Thanks for the comments. The treatment sequence in this study is to first administer radiotherapy, followed by endocrine therapy. A meta-analysis has indicated that concurrent radiotherapy with endocrine therapy does not significantly impact the incidence of radiation-induced toxicity or survival rates compared to a sequential approach [1]. In light of preclinical research suggesting enhanced therapeutic efficacy when radiotherapy is delivered prior to CDK4/6 inhibitors, we have opted to administer radiotherapy before the combination therapy of CDK4/6 inhibitors and hormone therapy [2].

In our study, we collected data on surgical wound recovery. All 12 patients had Class I incisions, which healed by primary intention. The wounds exhibited no signs of redness, swelling, exudate, or fat necrosis.

References:

- (1) Li Y F, Chang L, Li W H, et al. Radiotherapy concurrent versus sequential with endocrine therapy in breast cancer: A meta-analysis[J]. The Breast, 2016, 27: 93-98.
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Secondly, in the methodology, please describe the sample size estimation of this study and follow up details.

Thanks for pointing out this crucial omission. Sample size estimation for this study and follow-up details have been added in the methodology section. The section on sample size estimation has been revised to state in Statistical analysis: “This exploratory study involves 12 patients, with the sample size determined based on clinical considerations, not statistical factors (Lines 210-211).” The section on follow up has been revised to state in Procedures section “A 5-year follow-up is conducted every 3 months during the first 2 years, and every 6 months for the subsequent 3 years. Additionally, safety data are collected within 90 days after surgery for subjects who discontinue study treatment (Lines 169-172).”

Thirdly, in Table 1, the item HER2 expression, it's better to categorise HER2 into 0, 1+, 2+ and FISH-.

Thank you very much for pointing out this issue. The item HER2 expression in Table 1 has been revised from “negative, 1+, 2+ and FISH-” to “0, 1+, 2+ and FISH-”.

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